

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

CO5 – CONSTRAINT-BASED PROBLEM SOLVING
QUESTION 1 – RURAL BRANCH OFFICE NETWORK

1. Understanding the Problem

The company is opening a branch office in a remote rural location where high-speed wired Internet is unavailable. The network must support cloud-based applications, video conferencing, secure communication with headquarters, uninterrupted business operations, limited budget, and future expansion.

2. Constraints

- High-speed wired Internet is unavailable.
- Budget for connectivity is limited.
- Cloud applications require reliable Internet.
- Video conferencing requires sufficient bandwidth.
- Communication with headquarters must be secure.
- Business operations should continue even during connection failure.
- The network should support future expansion.

3. Recommended Internet Connection

I recommend **4G/5G wireless broadband as the primary Internet connection**. If 5G is unavailable, **4G LTE** can be used. If the budget permits, a secondary wireless or satellite connection can be used as a backup.

4. Networking Devices

- **4G/5G Router** – Provides Internet connectivity.
- **Firewall Router** – Protects the internal network.
- **Gigabit Switch** – Connects computers and other wired devices.
- **Wireless Access Point** – Provides Wi-Fi connectivity.
- **VPN Gateway** – Provides secure communication with headquarters.

5. Network Design

Internet → 4G/5G Router → Firewall → Switch → Computers/Access Points
A VPN tunnel can be established between the branch office and headquarters.

6. Security Mechanisms

- Use **VPN encryption** for communication with headquarters.
- Configure a **firewall** to block unauthorized traffic.
- Use strong passwords and **multi-factor authentication**.
- Use **WPA2/WPA3** for wireless security.
- Keep router and firewall software updated.
- Provide separate guest Wi-Fi.

7. Applying the Constraints

Constraint	Solution
No high-speed wired Internet	4G/5G wireless broadband
Limited budget	Cost-effective wireless equipment
Video conferencing	High-speed wireless connection
Secure communication	VPN + firewall
Uninterrupted operation	Backup Internet connection
Future scalability	Upgradeable router, switches and APs

8. Reliability and Scalability

A backup connection can automatically take over when the primary connection fails. This provides **failover** and reduces downtime.

The network can later be expanded by adding more switches, access points, and higher-capacity Internet services.

Conclusion

The proposed **4G/5G-based network with firewall, VPN, switch and wireless access points** provides a practical solution for the rural branch. It balances **bandwidth, reliability, security, cost and future scalability**, making it suitable for uninterrupted business operations.

QUESTION 2 – UNIVERSITY WIRELESS NETWORK FOR 3,000+ STUDENTS

1. Understanding the Problem

The university needs campus-wide wireless Internet for more than 3,000 students across classrooms, laboratories, libraries and hostels. The network must support many simultaneous users, minimize interference, provide secure access, and work within a limited access-point budget.

2. Constraints

- More than 3,000 students must be supported.
- Many users may connect simultaneously.
- Coverage is required across the campus.
- Signal interference must be minimized.
- Only a limited number of wireless access points can be deployed.
- The network must be secure.
- The solution must be cost-effective.

3. Access Point Placement

Access points should be placed according to **user density and building structure**, rather than simply placing them at equal distances.

More capacity should be provided in:

- Libraries
- Laboratories
- Large classrooms
- Hostels and common areas

A wireless site survey should be performed to identify dead zones and interference.

4. Frequency Band Selection

2.4 GHz

- Provides longer range.
- Better wall penetration.
- More interference.
- Fewer non-overlapping channels.

5 GHz

- Provides higher speed.
- Has more channels.
- Generally experiences less interference.
- Suitable for high-density areas.

Therefore, **5 GHz should be preferred in high-density areas**, while 2.4 GHz can be used where longer coverage is required.

5. Authentication and Security

Use **WPA2-Enterprise or WPA3-Enterprise** with centralized authentication.

Students can access the network using university credentials.

Separate networks can be created for:

- Students
- Staff
- Guests
- Administration

This improves security and prevents unauthorized access.

6. Network Design

Internet → Core Router → Core Switch → Distribution Switches → Wireless Access Points

A centralized wireless controller can be used to manage the access points.

7. Minimizing Interference

Interference can be reduced by:

- Proper channel planning.
- Using 5 GHz where possible.
- Adjusting access-point transmission power.
- Performing wireless site surveys.
- Avoiding unnecessary channel overlap.
- Using automatic channel selection where appropriate.

8. Balancing Constraints

Constraint	Solution
3,000+ users	High-capacity access points
Limited AP budget	Strategic AP placement
Signal interference	Channel planning
Large campus	Proper AP positioning
Security	WPA2/WPA3 Enterprise
Different users	Separate SSIDs/VLANs
High traffic	Prefer 5 GHz

Conclusion

An efficient campus wireless network should use **strategically placed high-capacity access points, mainly 5 GHz, centralized management, proper channel planning and enterprise authentication**. This provides good coverage and performance while reducing interference and controlling cost.

QUESTION 3 – SECURE NETWORK FOR FINANCIAL INSTITUTION

1. Understanding the Problem

The financial institution needs strong cybersecurity while employees must access sensitive customer information and online banking services. The network must follow strict security policies without significantly affecting performance or increasing operational costs.

2. Constraints

- Sensitive customer information must be protected.
- Online banking services must remain accessible.
- Strict security policies must be followed.
- Network performance should not be significantly reduced.
- Operational costs should be controlled.
- Unauthorized activities must be detected.

3. Network Segmentation

The network should be divided into different **VLANs/security zones**.

Suggested segmentation:

- 1. **Employee VLAN**
- 2. **Finance/Admin VLAN**
- 3. **Customer-Data VLAN**
- 4. **Server VLAN**
- 5. **Guest VLAN**
- 6. **Management VLAN**

Sensitive customer information should be placed in a restricted network segment. This prevents an attacker who compromises one segment from easily accessing the entire network.

4. Firewall Deployment

A **Next-Generation Firewall (NGFW)** should be deployed at the network boundary. The firewall should:

- Filter incoming and outgoing traffic.
- Block unauthorized connections.
- Control applications.
- Monitor suspicious traffic.
- Support VPN connections.
- Maintain security logs.

An internal firewall or additional security policies can protect sensitive servers.

5. Authentication

Use strong authentication methods such as:

- Username and password.
- **Multi-Factor Authentication (MFA).**
- Role-Based Access Control (RBAC).
- Strong password policies.

Employees should receive only the permissions required for their jobs.

6. Intrusion Detection

Deploy an **IDS/IPS**.

IDS	–	Intrusion	Detection	System:
Detects suspicious activities and generates alerts.				

IPS	–	Intrusion	Prevention	System:
Can automatically block detected malicious traffic.				

The system should continuously monitor network traffic and maintain security logs.

7. Secure Network Architecture

Internet → Firewall/NGFW → Core Switch → VLANs

The VLANs can include:

- Employee VLAN
- Server VLAN
- Customer-data VLAN
- Guest VLAN
- Management VLAN

Only authorized traffic should be allowed between the different segments.

8. Performance Considerations

Security should not unnecessarily slow down normal business operations.

To maintain performance:

- Use high-performance firewall hardware.

- Apply security rules based on risk.
- Monitor network traffic.
- Prioritize critical applications.
- Avoid unnecessary inspection of trusted traffic.
- Regularly review firewall rules.

9. Compliance and Cost

The architecture supports strict security policies through:

- Access control.
- Authentication.
- Network segmentation.
- Logging.
- Monitoring.
- Regular security updates.

To control costs, existing switches can be configured with **VLANs**, reducing the need for completely separate physical networks.

10. Constraint-Based Solution

Constraint	Solution
Sensitive customer data	Separate secure VLAN
Cybersecurity	Firewall + IDS/IPS
Strict policies	Access control + MFA
Performance	High-performance security devices
Low cost	VLAN-based segmentation
Unauthorized access	RBAC + MFA
Compliance	Logging and monitoring

Conclusion

The best solution is a **segmented network using VLANs, firewall/NGFW, MFA, role-based access control and IDS/IPS**. It protects sensitive financial information while maintaining network performance and controlling operational costs.

FINAL SUMMARY

For all three questions, the constraint-based approach is:

Identify Problem → Identify Constraints → Divide into Sub-problems → Select Network Design → Select Devices/Technologies → Apply Security → Consider Cost & Performance → Ensure Scalability → Give Conclusion

This structure directly addresses the assessment rubric: **understanding of constraints, problem decomposition, application of constraints, logical reasoning, feasibility, creativity and clarity of presentation.**